

4 There are 11 species of muntjac in Asia.

One of these is the Indian muntjac.

The photograph shows an Indian muntjac.



(Source: © blickwinkel / Alamy Stock Photo)

(a) The female Indian muntjac body cells have three pairs of chromosomes.

(i) Dividing cells can be taken from the body of a female Indian muntjac.

Draw one of these body cells showing the arrangement of chromosomes in the anaphase stage of mitosis.

(2)

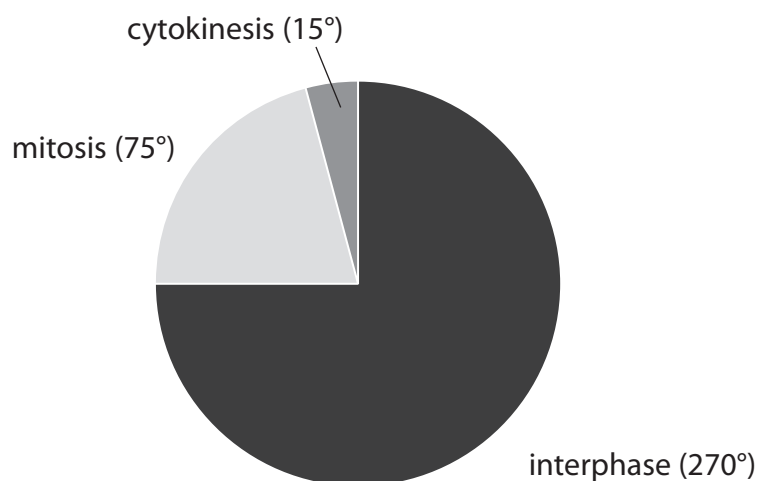
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- (ii) The diagram shows the relative proportions of time that a cell spends in each part of the cell cycle.



The cell was in interphase for nine hours.

The cell was in anaphase for 20 minutes.

Calculate the angle that would represent anaphase plotted on this pie chart.

(2)

Answer



P 6 9 4 9 9 A 0 9 3 2

(b) The diagram shows one of the chromosomes in an Indian muntjac cell.



(i) State what is meant by the term **locus**.

(1)

(ii) Explain how the chromosomes of an Indian muntjac egg cell could differ from those in a body cell.

(3)



(c) Chinese muntjac cells contain 23 pairs of chromosomes.

The photograph shows a Chinese muntjac.



(Source: © Design Pics Inc / Alamy Stock Photo)

The Chinese muntjac looks similar to the Indian muntjac. However, if they breed together, they produce offspring that are infertile.

Suggest why the offspring would be infertile.

(2)

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(Total for Question 4 = 10 marks)